



PLANNING PHASE – IT PROJECT

LocalAlert – a central missing-person application

Supervisor: **IT Project 700**Ms Mkhwanazi

Oratile Modise

402310679

Table of Contents

Introduction	2
2.1. Identification of need	3
2.1.1. The need and purpose	3
2.1.2. Who is it for?	3
2.1.3. Why It This Project Necessary?	4
2.1.4. Justification	4
2.2. Preliminary Investigation	5
2.2.1 The Project Goal	5
2.2.2 Project Scope	5
2.2.3 Potential Challenges	6
2.2.4 Preliminary Investigation Conclusion	8
2.3. Feasibility of Study	9
2.3.1. Technical feasibility	9
2.3.2. Operational feasibility	9
2.3.3. Economic feasibility	10
2.4. Project Planning	11
2.4.1 Project Planning Table	11
2.5. Project Scheduling.....	14
2.5.1 PERT Chart.....	14
2.5.2 Gantt Chart.....	15
2.6. Software Requirements Specification	16
2.6.1. Functional Requirements.....	16
2.6.2 Non-functional Requirements.....	17
2.7. Data Models.....	19
2.7.1 Entity Relationship Diagram.....	19
2.8. Bibliography	20

Introduction

Phase 2 of the LocalAlert app development focuses on the analysis and planning of the proposed mobile application. This phase translates the ideas, objectives and requirements identified during phase 1 into a more detailed plan for the development and implementation of the system. The purpose of this phase is to establish a clear understanding of what the LocalAlert application must do, who will use it, what resources are required, and whether the proposed system is technically, operationally and economically feasible. This phase will cover the identification of user needs, the system scope, project requirements, resources, risks, constraints and feasibility. It will also include project requirements and scheduling, the Software Requirements Specification (SRS), and the data model for the system. These activities will help how the system will work before development begins.

The application will provide several functions to support the missing-person alert process. These include Family Link, PIN-protected reporting, location-based alerts, automated alert escalation, community Spotted reporting and alert resolution. Family Link will allow users to store information about family members in advance, making it easier to create an alert to increase when the missing person has not been found within the set period. Community members will also be able to submit information when they see or identify a missing person. These functions are based on the main features identified in the Project Proposal.

2.1. Identification of need

The project aims to create a central system where South Africans can report about their loved ones missing, keep track and increase the reach of an alert.

2.1.1. The need and purpose

The purpose of LocalAlert is to provide a central mobile application where missing-person alerts can be created, managed and shared. Authorised users will be able to create an alert using information already stored in the system. This will reduce the time needed to prepare an alert during an emergency. The application will also allow verified alerts to be shared through existing social media platforms. This will increase the reach of an alert because users can share the same information with people who may not currently be using LocalAlert. It is needed to provide a faster, more organized and reliable way of communicating missing-person alerts to families, police, and member of the community.

2.1.2. Who is it for?

The main beneficiaries are families, police personnel and community members.

Families: Family administrators will be able to store family member information in advance and quickly create a missing-person alert when necessary. The Family Link system also allows family members to be connected within the application.

Police: Police personnel will be able to receive missing-person alerts and view information such as the child's details and location. The proposed police dashboard will also allow police to manage active alerts and spotted reports.

Community members: Community members will receive relevant alerts and will be able to report sightings. They can provide a location, directions and a photograph when reporting a sighting.

The wider community: People who receive a verified alert through WhatsApp or other socialmedia platforms can also become aware of the missing-person case and may provide useful information.

2.1.3. Why It This Project Necessary?

The project is necessary because missing-person information needs to reach the public quickly. The Phase 1 proposal explains that delays in informing the public can reduce the opportunity for people in the area to notice and report a missing child. It also identifies a gap in having a dedicated application for a verified, location-based missing-person alerts in South African communities.

There is also an opportunity to improve how existing communication platforms are used, WhatsApp and social media are already commonly used to share information, but information shared independently on these platforms may not organised or verified, location-based missing-person alerts in South African communities.

The use of mobile technology also makes the proposed solution practical, as communities have access to smartphones, approximately 127 million phones are active in South Africa.

2.1.4. Justification

LocalAlert provides a focused solution to a real communication problem identified in the Phase 1 proposal. Instead of depending only on separate WhatsApp messages, social-media posts, radio and other communication methods, the system will provide one platform for creating, managing, receiving and responding to the alerts.

The addition of external sharing makes the application more useful because verified alerts can be distributed beyond the LocalAlert user base. At the same time features such as false or accidental reports.

The project is therefore relevant because it combines a dedicated missing-person alert system with communication methods that communities already use. It provides a practical way to improve the speed and reach of missing-person information while keeping the original alert under the control of LocalAlert.

2.2. Preliminary Investigation

2.2.1 The Project Goal

The main goal of the LocalAlert project is to design, develop and test a mobile application that enables police, parents and community members to quickly send, receive and respond to verified missing-child alerts within their local area in South Africa. The system aims to provide the right information to the right people as quickly as possible through location-based alerts, Family Link, automated alert escalation and community-driven Spotted reporting.

The project will achieve this through several specific goals. First, LocalAlert will provide a Family Link system that allows a family administrator to create a family group and add family members. Relevant information such as the member's name, photograph, age and physical description can be prepared in advance so that it is immediately available when an emergency occurs.

Second, the system will provide PIN-protected one-button missing-person reporting. A family administrator will select the missing family member and confirm the report using a PIN before the alert is created. The alert will then be distributed to linked family members, the local police station and community members within an 800 m radius.

Third, LocalAlert will implement automated alert escalation. If the child is not found within 40 minutes, the alert radius will expand from 800 m to 20 km. Neighbouring police stations, and the wider community will then be notified.

Finally, the project will provide Community Spotted Reporting and Alert Resolution. Community members will be able to submit a sighting with their location, a short message and an optional photograph. Once the child has been confirmed safe, the family administrator can resolve the alert and notify users who received the original alert.

2.2.2 Project Scope

The scope of LocalAlert is to design, build and test a mobile application for missing-child alerts. The proposed application will include push notifications, location-based filtering, community tip reporting and a police administration dashboard.

The project will include the following major functions:

- Family Link for creating family groups and managing family-member profiles.
- ID-verified family profiles for storing information that can be used during a missing-person incident.
- PIN-protected one-button reporting for creating a missing-child alert.
- Location-based notifications to community members within the relevant search radius.
- Automated alert escalation from 800 m to 20 km after 40 minutes.
- Still Missing classification for alerts that remain unresolved after five days.
- Community Spotted Reporting with location, message and photograph options.
- Alert resolution when the missing child is found.
- Police administration dashboard for viewing active alerts, Spotted reports and escalation status.
- All 11 South African official languages will be supported, with users able to select their preferred language.

The proposed application will be developed as a cross-platform mobile application using React Native for Android and iOS. The proposed application will use React Native for the frontend, FastAPI with Python for the backend, PostgreSQL for the database, and JWT tokens for authentication.

The project will not include full integration with SAPS computer systems and will not proceed beyond a pilot testing environment.

2.2.3 Potential Challenges

Several challenges may arise during the design, development and testing of LocalAlert.

1. Integration with existing police systems

The proposed system will not include full integration with SAPS computer systems. This means the project must operate as a standalone application during the pilot stage rather than directly exchanging information with existing police databases.

2. Reliable location-based alerts

LocalAlert depends on location information to determine which community members should receive an alert. The system therefore needs to accurately identify the relevant search area while avoiding unnecessary notifications outside that area.

3. Alert escalation

The automated expansion from 800 m to 20 km after 40 minutes must operate correctly. The project must ensure that the correct users and neighbouring police stations are notified when an alert escalates.

4. False or accidental reports

A major concern is preventing unauthorised or accidental missing-person reports. The PINprotected reporting mechanism is therefore included to ensure that an alert is deliberately confirmed before it is distributed.

5. Data security and privacy

The system will handle sensitive information about children and families, including photographs, physical descriptions and identification information. Protecting this information and ensuring that only appropriate users can access it will therefore be an important development challenge.

6. Internet connectivity

The application relies on mobile connectivity to distribute alerts and receive community responses. This is an important consideration because the proposal specifically identifies different levels of internet access among South African communities as a factor that the system should take into account.

7. User adoption and usability

LocalAlert must be simple enough for family administrators, community members and police personnel to use during a stressful missing-child situation. The project therefore needs to focus on clear interfaces and straightforward reporting procedures.

8. Testing with appropriate users

The proposal requires testing with community members, family administrators and police, focusing on alert reliability, escalation timing and the ease of the Spotted reporting process. The project plans to conduct User Acceptance Testing with at least ten participants.

2.2.4 Preliminary Investigation Conclusion

The preliminary investigation indicates that LocalAlert addresses a specific problem: the absence of a dedicated South African mobile application for quickly distributing verified, location-based missing-child alerts to relevant community members and police. Current methods such as social media, WhatsApp and radio announcements are described in the proposal as unreliable, unorganised and unable to effectively target people near the location where the child went missing.

The investigation also confirms that the proposed system has a clearly defined scope and identifiable technical challenges. The project will therefore proceed to the Feasibility Study, where the proposed LocalAlert solution will be assessed in terms of its technical, operational and economic feasibility as required by the project guidelines.

2.3. Feasibility of Study

2.3.1. Technical feasibility

The technical resources required to design, develop, and implement LocalAlert are available and accessible. I have, at my disposal, a Windows laptop for research and development, and a mobile phone for testing. The tools, frameworks, and platforms that will be used are as follows: React Native with Expo will be used to build the cross-platform mobile application for both Android and iOS. Fast API with Python will be used to build the backend Rest API. PostgreSQL will be used and the relational data to store all the application data. JWT tokens will be used for use authentication and authorization. Figma will be used to design the user interface wireframes and screen mockups. Postman will be used to test and document all API endpoints. GitHub will be used for version control and source code management. All of the above tools are free and do not require any paid subscription for the purpose of this academic prototype. Prior knowledge of React Native, Python, and database design acquired through the BSc IT programme further supports the technical viability of this project. Therefore, LocalAlert is technically feasible given the available tools, skills, and resources at my disposal.

2.3.2. Operational feasibility

I am the only person responsible for all aspects of this project, including planning, UI/UX design, frontend, backend development, testing, and documentation. To ensure that the project is completed on time, a detailed project plan and development schedule has been created, outlining tasks and deadlines for each phase of the project. Figma will be used to design all application screens before development begins. Visual Studio Code will be used as the primary code editor. GitHub will be used to track all code changes and maintain version history throughout the development process. Testing will be conducted using Expo Go on a physical mobile device during development. User Acceptance Testing will be conducted within a minimum of ten participants, including family members and friends who will serve as ad hoc testers. Since this is a solo project, time management is critical. I will prioritize core features first and ensure that the most important functionality is completed before the submission

deadline. Therefore, LocalAlert is operationally feasible provided the development schedule is followed consistently and time is managed effectively.

2.3.3. Economic feasibility

The cost of designing, developing and implementing LocalAlert as an academic project is minimal. All tools, frameworks and platforms required for this project are available free of charge. React Native and Expo are open source and free to use. FastAPI and python are open source and free to use. PostgreSQL is an open-source relational database management system available at no cost. VS Code is a free code editor. GitHub offers free repositories for individual developers. Figma offers a free tier that is sufficient for designing all screens and wireframes for this project. Postman offers a free tier that is sufficient for testing and documenting all API endpoints. No cloud hosting costs are incurred during development and testing, as the FastAPI backend will run locally on my machine. File uploads, such as ID document photos and Spotted report photos, will be stored on the local filesystem, eliminating the need for any paid storage service. Should LocalAlert be scaled beyond academic prototype in the future, costs would include cloud hosting services such as Google Cloud Run, a managed PostgreSQL hosting service, and app store submission fees. The Google Play Store requires a once-off registration fee of approximately R450, while the Apple App Store required an annual fee of approximately R1 720. However, these costs are not applicable to this academic project. Therefore, LocalAlert is economically feasible as all required tools and resources are available at no cost to the developer.

2.4. Project Planning

The project planning stage establishes a structured plan for designing, developing, testing and preparing the LocalAlert application. The project will follow the Systems Development Life Cycle (SDLC), with each phase producing specific outputs that contribute towards the final system. The planning process will ensure that the project remains within its defined scope, available resources and academic timeframe.

2.4.1 Project Planning Table

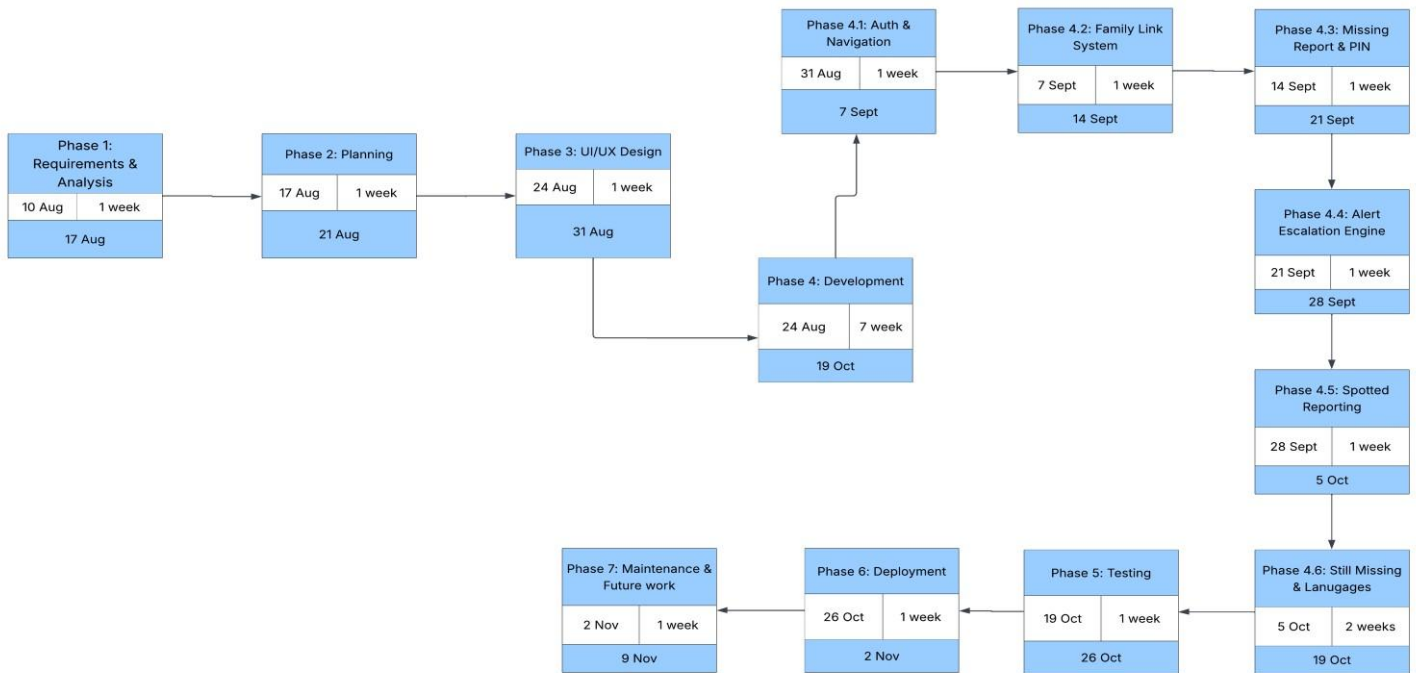
Week	Project Phase	Expected Deliverables	Main Activities
Week 1	Phase 1: Project Proposal	Project proposal describing the implementation language/technologies and selected SDLC model.	Define the problem of delayed communication during missing person cases. Define the purpose, objectives and scope of LocalAlert. Identify the proposed implementation technologies and SDLC approach
Week 2	Phase 2: Planning	• Identification of need • Preliminary investigation • Feasibility study • Project planning • PERT chart • Gantt chart • Software Requirements Specification (SRS) • Data models	Identify the need for LocalAlert. Conduct a preliminary investigation into existing missing-person reporting and alert methods. Conduct technical, operational, and economic feasibility studies. Develop the Software Requirements Specification. Develop initial data models.

Week 3	Phase 3: Analysis	Information-gathering results, existing-system analysis, weaknesses of the current system, functional requirements, non-functional requirements and proposed-system data models.	Gather information using appropriate methods such as interviews and observation. Analyse existing missing-person reporting methods. Identify weaknesses of the current system. Analyse the proposed LocalAlert system and define functional and non-functional requirements. Develop data models for the proposed system.
Week	Phase 4: System Design	System design, architectural diagrams, class diagrams, physical design, pseudocode, interface design and security/backup design.	Design the proposed LocalAlert system. Develop the software, hardware and network architecture. Develop class diagrams and database designs. Design program logic and pseudocode. Design security and backup design.

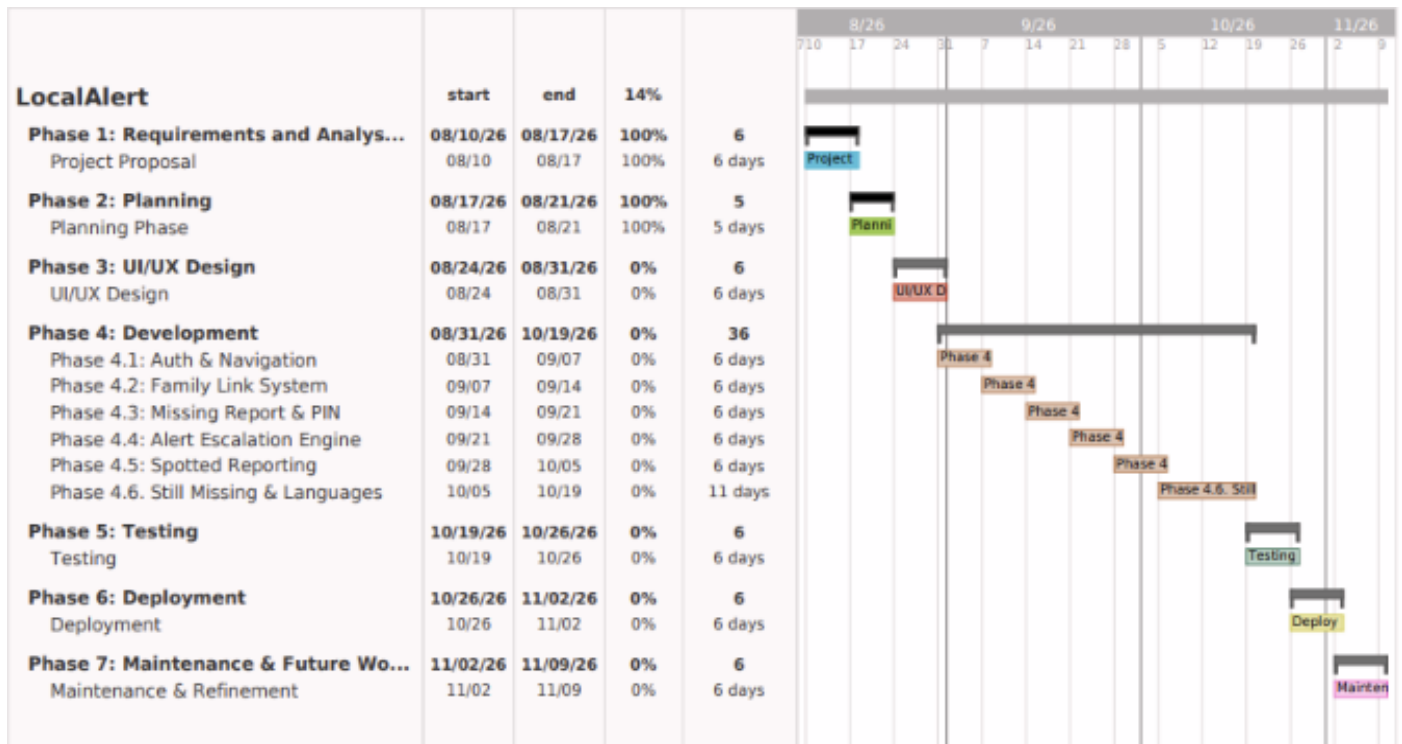
	Phase 5: Implementation	• Completed application • Source code • Testing results • System test cases • Evaluation of results • Software installation	Develop the LocalAlert application using the selected programming technologies. Implement the database, authentication, Family Link, missing-person reporting, alert notification, location-based escalation and Spotted reporting. Conduct unit and system testing, evaluate test
			results and install the application for demonstration.

2.5. Project Scheduling

2.5.1 PERT Chart



2.5.2 Gantt Chart



2.6. Software Requirements Specification

2.6.1. Functional Requirements

1. The system shall allow users to register an account using their full name, email address, phone number, password, and role (community member, family admin, or police officer).
2. The system shall allow users to log in and log out of the application using JWT tokenbased authentication.
3. The system shall allow a family admin to create a family group and add verified profiles of their family members, including their full name, date of birth, gender, physical description, profile photo, and ID document photo.
4. The system shall require a family member profile to have an uploaded ID document photo before it can be used in a missing alert, to reduce false reports.
5. The system shall allow a family admin to trigger a missing alert by selecting a verified family member profile and confirming the report with a 4-digit PIN.
6. The system shall automatically notify all linked family members, the local police station, and community members within an 800m radius when a missing alert is created.
7. The system shall automatically expand the alert radius from 800m to 20km after 40 minutes if the missing person has not been found and shall notify all linked family members and neighbouring police stations of the escalation.
8. The system shall allow any community member who has received an alert to submit a Spotted report containing at least one of the following: their live GPS location, a written directions message, or a photo as proof.
9. The system shall send all Spotted reports directly to the linked family members and the police.
10. The system shall allow a family admin to resolve a missing alert by confirming with their 4-digit PIN, after which a found notification shall be sent to all users who received the original alert.

11. The system shall automatically move any active alert that has not been resolved within 5 days to a Still Missing category, where it remains visible and can still receive Spotted reports.
12. The system shall support all 11 South African official languages, allowing users to select their preferred language on first launch and change it at any time through the settings screen.
13. The system shall store all uploaded files, including ID document photos and Spotted report photos, securely on the server and restrict access to ID documents to the family admin and verified police users only.
14. The system shall present a privacy notice to users at registration and request their consent for data collection in compliance with the Protection of Personal Information Act (POPIA).

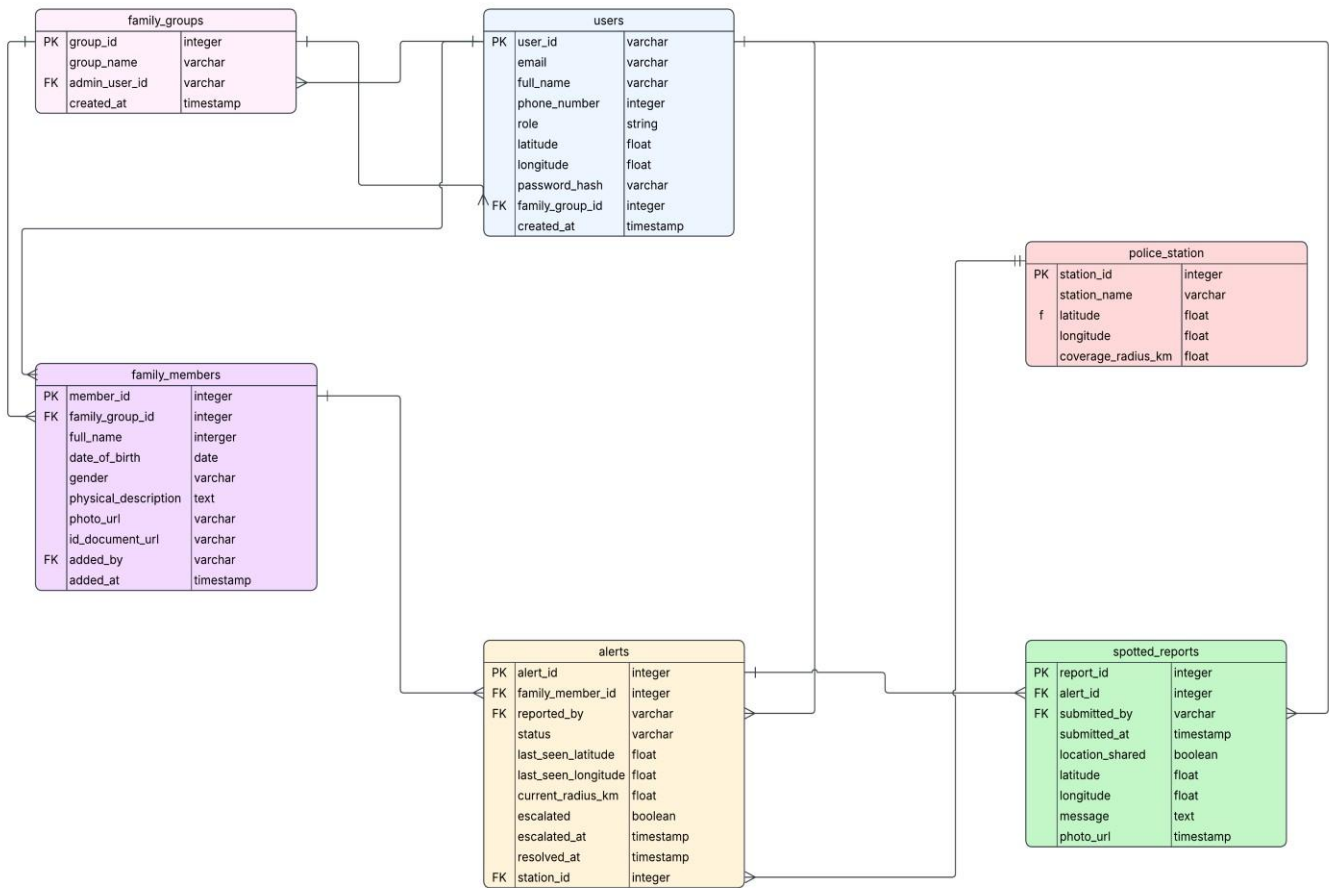
2.6.2 Non-functional Requirements

1. The system shall load the main screen within 3 seconds on a modern Android or iOS device with a stable internet connection.
2. The system shall respond to all API requests within 2 seconds under normal operating conditions.
3. The system shall hash all user passwords and family admin PINs using bcrypt before storing them in the database. Passwords and PINs shall never be stored or transmitted as plain text.
4. The system shall authenticate all API requests using JWT tokens. Any request without a valid token shall be rejected with a 401 Unauthorized response.
5. The system shall enforce role-based access control, ensuring that only a family admin can create alerts, resolve alerts, and manage family member profiles.
6. The system shall be compatible with both Android and iOS devices through the React Native and Expo cross-platform framework.
7. The system shall require an active internet connection to function. Offline mode is not supported in version 1.0 of the application.

8. The system shall ensure that ID document photos are only accessible to the family admin who uploaded them and verified police users. They shall not be visible to community members or other users.
9. The system shall be built with a clean, well-documented codebase using a structured folder system, making it easy to maintain, update, and extend in future versions.
10. The system shall comply with the Protection of Personal Information Act (POPIA) by collecting only the data necessary for the application to function and by not sharing user data with any third parties.
11. The system shall use HTTPS for all communication between the mobile application and the backend API, ensuring that all data in transit is encrypted.
12. The system shall be able to filter community members within a specified GPS radius using the Haversine formula, ensuring that alerts are only sent to users within the relevant geographic area.

2.7. Data Models

2.7.1 Entity Relationship Diagram



2.8. Bibliography

- Aliyu, M. (2023). *Mobile application development with React Native: A practical guide*, 14 June 2023. [Online]. Available at: <https://medium.com/@maliyuabubakar/mobileapplication-development-with-react-native-a-practical-guide> [Accessed 22 August 2026].
- Axiomq. (2024). *Mobile app development process: Common challenges and their solutions*, 13 September 2024. [Online]. Available at: <https://axiomq.com/blog/mobile-app-development-process-common-challengesand-their-solutions/> [Accessed 22 August 2026].
- Beram, S. (2025). *How to conduct a feasibility study: Step-by-step guide with examples*, 2 April 2025. [Online]. Available at: <https://blog.logrocket.com/productmanagement/how-to-conduct-feasibility-study/> [Accessed 22 August 2026].
- Boyle, M. (2025). *6 key things to include in your app terms and conditions*, 8 January 2025. [Online]. Available at: <https://sprintlaw.co.uk/articles/6-things-to-include-appterms-conditions/> [Accessed 22 August 2026].
- Cedeño, D.E. (2021). *10 recommendations to make your app accessible*, 28 December 2021. [Online]. Available at: <https://www.insuit.net/10-recommendations-to-makeyour-app-accessible/> [Accessed 22 August 2026].
- Information Regulator South Africa. (2022). *Protection of Personal Information Act (POPIA): Guidance for organisations*, 1 July 2022. [Online]. Available at: <https://infoeregulator.org.za> [Accessed 22 August 2026].
- Kanal, B. (2024). *Audience targeting tips to reach the right users for your mobile app*, 22 August 2024. [Online]. Available at: <https://www.shyftup.com/blog/audientargeting-tips-to-reach-the-right-users-for-your-mobile-app/> [Accessed 22 August 2026].
- Medium. (2024). *Common challenges in mobile app development and how to overcome them*, 12 July 2024. [Online]. Available at: <https://medium.com/@emperorbrains/common-challenges-in-mobile->

[appdevelopment-and-how-to-overcome-them-0bfba9acdfb2](#) [Accessed 22 August 2026].

- National Planning Commission. (2012). *National Development Plan 2030: Our future — make it work*. Pretoria: The Presidency of the Republic of South Africa.
- Nkosi, B. (2023). *Digital public safety gaps in South African communities*. South African Journal of Information Technology, 9(1), pp. 18–34.
- PostgreSQL Global Development Group. (2024). *PostgreSQL 16 documentation*, 2024. [Online]. Available at: <https://www.postgresql.org/docs/> [Accessed 22 August 2026].
- Singh, S. (2025). *11 biggest challenges in mobile app development*, 10 February 2025. [Online]. Available at: <https://appinventiv.com/blog/mobile-app-developmentchallenges/> [Accessed 22 August 2026].
- Smith, A. and Dlamini, P. (2022). *Critical hours: Response time and child recovery rates in South African missing persons cases*. African Safety Research Quarterly, 6(3), pp. 102–118.
- South African Police Service. (2023). *Annual report 2022/2023*. Pretoria: South African Police Service.
- Statista. (2024). *Mobile internet penetration rate in South Africa 2024*, 2024. [Online]. Available at: <https://www.statista.com> [Accessed 22 August 2026].